

A-Door-Able Homes

Pads, Palaces, and Premiums

Luxury

Evidence

The King County data that justifies your expensive taste

(and the three recommendations we tested until they broke)

21,597 sales · May 2014 – May 2015 · King County, WA

THE BRIEF

Jennifer Montgomery wants a lot of things at once.

WHAT SHE ASKED FOR

- High budget, wants to show off
- Waterfront access
- Recently renovated
- High grade and a real view
- Buy within a month
- Resell within a year

WHAT THAT MEANS IN THE DATA

`price >= $887K (90th percentile)`

`waterfront == 1`

`year_renovated > 0`

`grade >= 10 and view >= 3`

`condition >= 3`

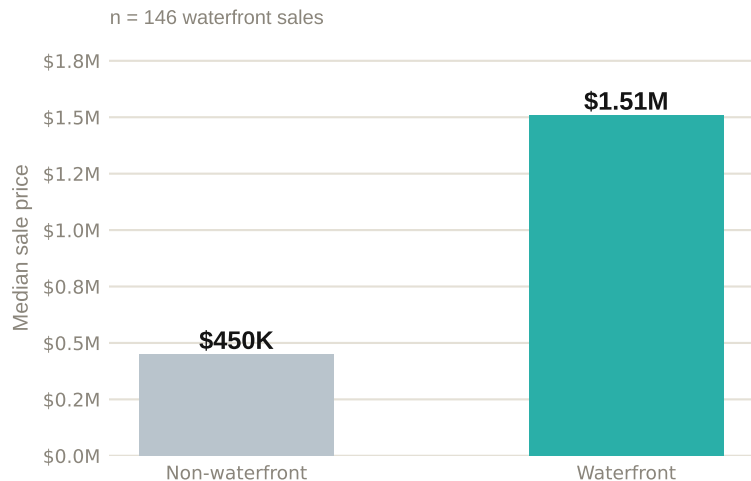
5 properties survive all six.

The scarcity is the finding.

Six hard requirements leave a handful of houses out of 21,597.
Before anything else, the client needs to know which requirement she would most like to relax.

WATERFRONT IS NOT A FEATURE. IT IS A SEPARATE MARKET.

Hypothesis H3 - confirmed, and by a wider margin than any other feature in the dataset.



THE PREMIUM

Waterfront homes sell at a median of \$1.51M against \$450K — 3.4x.

THE SCARCITY

Only 146 of 21,597 sales are waterfront. Roughly 0.7% of the market.

WHY IT IS NOT JUST 'A VIEW'

Grade tracks it too: waterfront homes cluster in the top build grades. The market treats shoreline as a different product, not a nicer version of the same one.

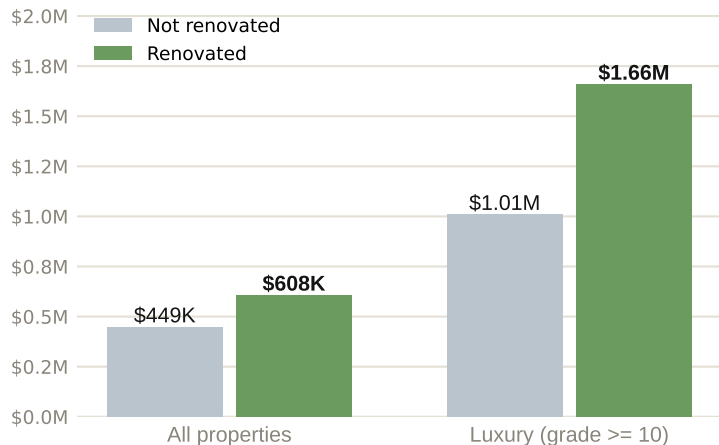
H3 CONFIRMED

FOR THE CLIENT

Worth paying for - but it is also the requirement that shrinks the candidate list fastest, from thousands to dozens.

RENOVATION IS THE ONE THAT SURVIVES EVERY CHECK.

Hypothesis H5 - the only price finding here that still holds once confounders are controlled for.



THE UPLIFT

+35% market-wide, and
+64% inside the luxury segment.

THE OBVIOUS OBJECTION

Renovated homes are also older, bigger
and better located. So is the premium
real, or is it just size in disguise?

THE CONTROL

Split the market into living-area bands
and the premium survives in every band
above 1,500 sqft - though it vanishes

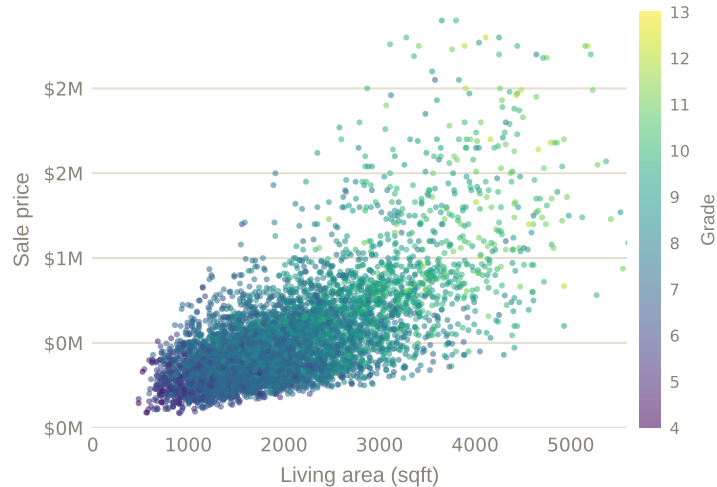
H5 CONFIRMED

THE HONEST WORDING

Renovated homes sell for more. That is
not the same claim as renovating adding

SIZE AND GRADE DO THE WORK.

Hypothesis H4 - living area is the single strongest predictor of price anywhere in the data.



THE DRIVERS

sqft_living $r = 0.70$
grade $r = 0.67$
sqft_above $r = 0.61$
sqft_living15 $r = 0.59$
bathrooms $r = 0.53$

Every strong predictor is a measure of size or build quality. None is a measure of timing.

THE VALUE ANGLE

\$450k - \$1M homes

H4 CONFIRMED

largest homes drop back below both.

Size is bought at a discount at the top.

TESTED, THEN REJECTED

We looked for a cheapest day of the month to buy. We found one - then tried to break it.

THE CLAIM

Day 10 has the lowest median luxury price (\$1.26M).

THE PROBLEM

That median rests on 7 sales. The typical day-of-month bucket holds 7.

THE TEST

Shuffling the day labels 1,000 times reproduces a gap this large 75% of the time.

THE REALITY

Across all 21,597 sales day 10 ranks 28 of 31 - among the dearest. The pattern reverses at scale.

H7 REJECTED · no day-of-month advice

p = 0.75

Permutation test, corrected for having picked the best of 31 days. Indistinguishable from noise.

"We could have sold you a lucky date. We would rather sell you a house."

SEASONALITY IS REAL. IT IS ALSO TINY.

The same test on the month of sale - where, unlike the day, a real effect does exist.

THE CLAIM

Sep beats Aug by 43%
(\$2.12M vs \$1.49M).

THE PROBLEM

18 sales versus 28. Permutation $p = 0.44$.

AT SCALE

Market-wide a month effect IS detectable —
but the same two months differ by
only 1.8%.

THE LESSON

Significance answers 'is it real?'. Effect
size answers 'do I care?'. Here the answers
are yes and no.

H6 REJECTED · it predicted January anyway

1.8%

The real Sep-vs-Aug gap once all
21,597 sales are used, instead of
the 240 in the luxury pool.

*"A 2% seasonal edge does not
survive a 6% transaction fee."*

THE RESALE PLAN NEEDS A RETHINK

The client wants to resell inside a year. Repeat sales answer that question directly.

THE DATA

176 houses here sold twice. 5 sit inside the client's own price and grade segment.

THE RETURNS

+7.1% +6.5% +1.9% +1.4% -0.8%
Median: +1.9% over 176 days.

THE COSTS

Round-trip transaction costs on a property sale: 6-10%.

THE ADVICE

Buy to hold, or revise the target. The one-year flip does not clear its own costs.

BUY TO HOLD · the flip does not pay

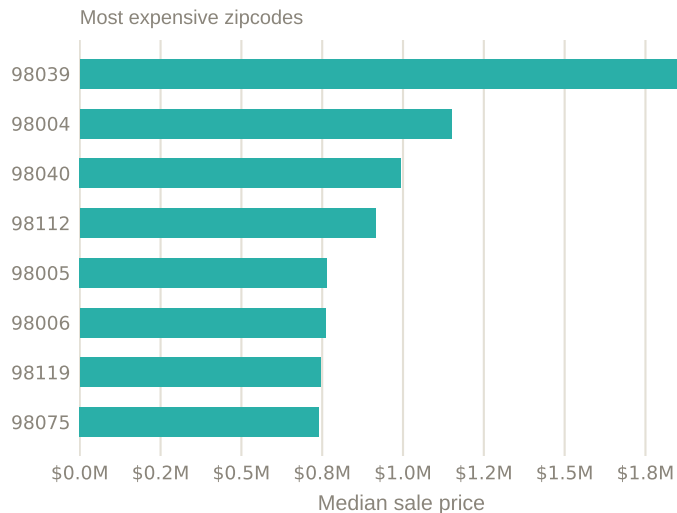
+1.9%

Median return on comparable resales in her own segment, before 6-10% in costs.

"The market-wide figure says +54%. That is people flipping starter homes, not waterfront."

WHERE THE MONEY ACTUALLY SITS.

Drawn from all 240 luxury sales, not inferred from a five-house shortlist.



THE PRICIEST

98039 median \$1.90M

98004 median \$1.15M

98040 median \$994K

THE DEEPEST MARKET

Most luxury transactions:

98006 (40) 98040 (24) 98199 (18)

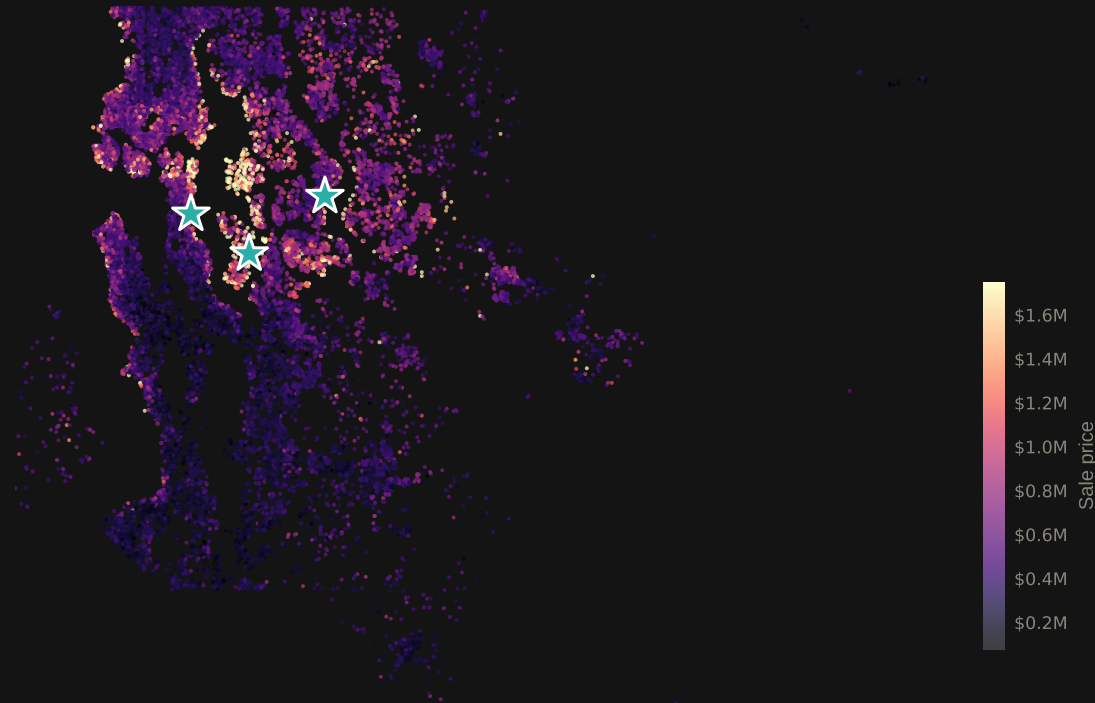
WHY IT MATTERS

Prestige and depth are not the same place. The dearest zipcode is thin.

GEOGRAPHIC INSIGHT

THE PRICE MAP

*Every one of the 21,597 sales, placed by latitude and longitude and coloured by price.
Nobody drew a boundary here - the county's shape is the transactions themselves.*



THE SPINE OF VALUE

The bright band running north-south is the Lake Washington shoreline - Medina, Bellevue, Mercer Island. Price falls away with distance from the water in almost every direction.

THE SOUTH IS DIFFERENT

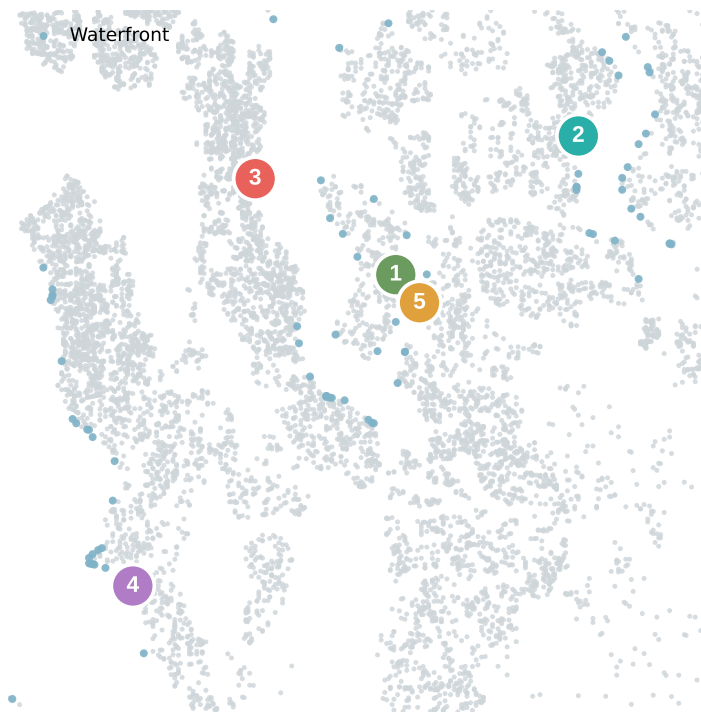
The dark mass in the south of the county is a larger, cheaper market. It is not short of houses; it is short of the attributes this client wants.

THE STARS

The three recommended properties. They sit exactly where the map says value concentrates.

WHERE THE SHORTLIST SITS

Zoomed to cover every property that survived all six requirements.



THE SURVIVORS

1	\$4.67M • 98040
	house_id 1924059029 grade 12
2	\$3.30M • 98008
	house_id 3625059152 grade 11
3	\$3.60M • 98144
	house_id 4114601570 grade 10
4	\$1.64M • 98166
	house_id 6117502230 grade 12
5	\$1.80M • 98006
	house_id 3343302110 grade 10

They are not clustered.

The survivors are scattered across five separate submarkets. There is no single neighbourhood to shop in - each of these is a one-off, which is precisely what makes the brief hard to satisfy.

THE SHORTLIST

Ranked by how far each stands out from its own neighbourhood.

house_id 1924059029

\$4.67M

Zipcode	98040
Grade	12 of 13
View	4 of 4
Renovated	2009
Living area	9,640 sqft
Premium score	20.65

house_id 3625059152

\$3.30M

Zipcode	98008
Grade	11 of 13
View	4 of 4
Renovated	1987
Living area	4,220 sqft
Premium score	5.46

house_id 4114601570

\$3.60M

Zipcode	98144
Grade	10 of 13
View	4 of 4
Renovated	2002
Living area	5,020 sqft
Premium score	4.84

Plus 10 more the filters cannot confirm.

Those properties meet every other requirement, but their renovation year is missing from the record. Filling that gap with a zero would quietly delete them — so they are reported separately rather than dropped.

THE RESULT

Seven hypotheses went in. Five survived, two did not.

H1 Missing waterfront data means 'not waterfront'

CONFIRMED

H2 Missing renovation year means 'never renovated'

CONFIRMED

H3 Waterfront commands a premium (3.4x)

CONFIRMED

H4 Living area drives price ($r = 0.70$)

CONFIRMED

H5 Renovation lifts price (+35%, +64% luxury)

CONFIRMED

H6 Luxury prices peak in a given month ($p = 0.44$)

REJECTED

H7 Day of month predicts price ($p = 0.75$)

REJECTED

WHAT THE CLIENT GETS

BUY

5 properties clear every requirement, plus 10 the records cannot confirm.

PAY FOR

Waterfront and a renovation. Both are priced in, and both hold up under testing.

IGNORE

Timing. Neither the month nor the day survived a significance test.

RECONSIDER

The one-year flip. Comparable resales returned +1.9% before costs.

The two rejections are the point.

Any analysis can find a pattern in 240 rows. Testing whether that pattern survives contact with the full 21,597 sales is what separates a finding from a coincidence - and it cost this deck two of its most quotable slides.

WHAT WE WOULD CHECK NEXT

KNOWN LIMITATIONS

Waterfront gaps were imputed from neighbours; low recall at a 0.7% base rate.

Renovation gaps filled with zero, which removes otherwise-qualifying houses.

176 repeat sales are double-counted in every market-wide median.

One county, one 13-month window.

Holding periods are capped by the window, so 'all resold within a year' is an artifact.

HOW THIS WAS TESTED

21,597 sales joined from two PostgreSQL tables.

Every gap explained before it was filled.

Permutation tests corrected for picking the most extreme of many buckets.

Two of our own hypotheses rejected.

Findings that do not survive testing are worth more than findings that were never tested.



THANK YOU!

Questions, challenges and better hypotheses all welcome.